QR-004 Mold sealing and release, in order

Doc ID	QR-004
Title	Mold sealing and release, in order
Parent	CS-004 §5; CS-004 §7.1; CS-004 §7.2; CS-004 §7.3; CS-004 §7.4; CS-004 §7.5; CS-004 §8
Checked against	CS-004 Rev E; CS-004 Rev E; CS-004 Rev E; CS-004 Rev E; CS-004 Rev E; CS-004 Rev E; CS-004 Rev E

Derived from its parent standard. If this and the parent disagree, the parent is right.

1. Mold sealing and release, in order

Clean, inspect, seal with XCR, wait the 12 h hold after the last coat, wax (4 coats minimum on a new mold), wait 1 h, PVA (2 coats), then label the mold and set it to Ready for layup (CS-004 §7.1 to §7.5).

Also called: mold prep, mould prep, seal and release, sealing, release, mold sealing order.

1. Clean the machined mold: dust off, then wipe with IPA on microfiber (CS-004 §7.1).
2. Inspect: every scribe line and datum crisp; photograph the dry mold; fill gouges per CS-003 (CS-004 §7.2).
3. Seal with XCR, 150 g or less per batch at 100:41 by weight (CS-004 §7.3).
4. Overcoat while the first coat is firm but tacky, checked with a fingernail (CS-004 §7.3 step 6).
5. Wait 12 h after the last coat, the FEB hold (CS-004 §7.3 step 7).
6. Wax with Formula Five: 4 coats minimum on a new mold, buffing between coats (CS-004 §7.4 step 1).
7. Wait at least 1 h after the last wax coat (CS-004 §7.4 step 2).
8. PVA with Partall #10: 2 coats, each dry before the next (CS-004 §7.4 step 3).
9. Record the seal on the mold and in the “Seal and release mold” note (CS-004 §7.4 step 5).
10. Label the back face and set Stage to Ready for layup (CS-004 §7.5).

QR-004 §2 to §8 cover each step in turn.

2. Cleaning a mold before sealing: IPA, never acetone

Clean a machined mold with 91% isopropyl alcohol on a microfiber cloth. Never acetone on tooling board, which attacks the board, and never fibrous shop towels, which leave strings in the surface (CS-004 §7.1).

Also called: cleaning, wipe down, IPA, isopropyl, rubbing alcohol, acetone, solvent, dust, mold cleaner.

1. Blow all dust off the mold with compressed air, including pockets and scribe lines (CS-004 §7.1 step 1).
2. Wipe the whole surface with an IPA-dampened microfiber cloth (CS-004 §7.1 step 2).
3. Repeat with a clean face of the cloth until it comes away clean.
4. From here on, handle the mold with gloves only. Skin oils contaminate the surface (CS-004 §7.1 step 3).

Which IPA: the 91% isopropyl bottle, which the app records as “Isopropyl alcohol 91%”. Never the 70% rubbing alcohol (CS-004 §5).

Acetone and fibrous towels are both failure modes the team documented in SN4 (CS-004 §7.1).

A re-used mold that is already sealed is cleaned with Formula Five Clean ’N Glaze, then gets 1–2 wax coats and PVA (CS-004 §7.4 step 4; QR-004 §7).

3. Mixing XCR to seal a mold: batch size and ratio

Mix XCR 150 g or less at a time, at 100:41 resin to hardener by weight. Pot life at that size is 8 minutes at 20 °C and 4 at 30 °C, and a bigger batch is shorter-lived and hotter (CS-004 §7.3 step 2).

Also called: XCR, sealer, mold sealer, seal coat, coating resin, batch size, mixing XCR.

1. Check the room is 15–30 °C and can stay at 20 °C or warmer through the cure (CS-004 §7.3 step 1).
2. Check both jugs are inside 15–30 °C. On a cold night, bring them in from storage first.
3. Avoid a humid room. Do not seal outdoors on cold evenings: the datasheet rates XCR for 15–30 °C only (CS-004 §7.3 step 1).
4. Weigh XCR at 100:41 resin to hardener by weight, 150 g or less (CS-004 §7.3 steps 2 and 3).
5. Mix at least 2 min, scraping walls and base.
6. Double-pot: transfer to a second cup and remix with a fresh stick.
7. Write each batch mass on the work order.

A 150 g batch covers roughly a medium mold section per coat. Mix again rather than mixing big (CS-004 §7.3 step 2).

Anyone who has had a skin reaction to epoxy shall not mix or brush XCR, and tells a lead (CS-004 §6). Nitrile gloves and safety glasses with side shields whenever XCR is open (CS-004 §6). Unmixed streaks from the cup walls are the classic soft-spot cause (CS-004 §7.3 step 3).

4. How many XCR coats, and when to put on the second coat (fingernail test)

Usually two coats on 30 lb board, often one on 60 lb, two on 45 lb (team rule); a mixed-grade mold seals like its softest board (CS-004 §7.3 step 4). Overcoat when the first coat is firm but tacky (fingernail dents it), about 2 h 50 at 20 °C (CS-004 §7.3 step 6).

Also called: second coat, recoat, overcoat, fingernail test, firm but tacky, B-stage, tack-free, how many coats.

How many coats: 30 lb board usually needs a second coat, and 60 lb board often seals in one. Plan two coats on 45 lb board until the team has a record of it sealing in one (team rule). A mold cut from mixed grades seals like its softest board, so you should plan two coats if any of it is 30 or 45 lb. The mold's "Board density, min" and "Board density, max" fields show the range (CS-004 §7.3 step 4).

1. Brush a thin, even coat over the working surface and flanges (CS-004 §7.3 step 4).
2. Work resin out of corners and fillets. A pool in a crevice is the Duratec failure.
3. Watch the scribe lines. If one floods, drag the brush along it before the resin gels (CS-004 §7.3 step 5).
4. Press a fingernail into the first coat on the flange. Firm but tacky means overcoat now; it bonds with no prep (CS-004 §7.3 step 6).

At 30 °C the window comes at about 1 h 30. Either time is an estimate, and the fingernail decides (CS-004 §7.3 step 6). Missed the window and the coat is tack-free: QR-004 §5.

5. First coat went tack-free? Sanding XCR before the next coat

If the first coat has gone tack-free (coming back another day always means tack-free), wait at least 9 h from application, then key it with 120 grit before recoating (CS-004 §7.3 step 6). Firm-but-tacky XCR needs no sanding.

Also called: sand XCR, sanding XCR, key, scuff, 120 grit, missed the overcoat window, recoat next day, tack-free, second coat next day.

1. Wait at least 9 h from application (CS-004 §7.3 step 6).
2. Key the surface with 120 grit.
3. Blow off the dust.
4. Wipe with IPA.
5. Recoat, mixing 150 g or less at 100:41 by weight (CS-004 §7.3 steps 2 and 3).

6. How long after XCR before wax? The 12 h hold

Wait 12 h after the last XCR coat before the wax and PVA go on. That is the team's FEB hold; the datasheet initial cure is 8 h 30 at 20 °C, handleable at about 9 h (CS-004 §7.3 step 7). The app does not lock this hold, so watch it yourself.

Also called: XCR cure time, seal cure, how long to cure the sealer, 12 hours, FEB hold, when can I wax.

1. Write the last-coat time in the note on the work order's "Seal and release mold" step (CS-004 §7.3 step 7).
2. Wait 12 h from that time.
3. Press a fingernail into the flange, not the part surface. Pass is no mark (CS-004 §8).
4. Only if a run or nib formed: denib it locally with 800 grit or finer (a team value), then wipe with IPA (CS-004 §7.3 step 7).
5. Keep water off the sealed surface for 24 h after application (CS-004 §7.3 step 8).

Check the scribe lines against the photo of the dry mold: all of them identifiable, and no glossy pools in fillets or corners under raking light (CS-004 §8).

7. Wax then PVA: coats, and how long each must dry

New mold: Formula Five wax, 4 coats minimum, buffing between coats, a wait of at least 1 h, then Partall #10 PVA, 2 coats, each dry before the next, about 15–45 min a coat (CS-004 §7.4; `03 Datasheets/PARTALL-Film-10.pdf`).

Also called: release, release agent, mold release, wax, Formula Five, PVA, Partall, Clean 'N Glaze, mold cleaner, re-wax.

1. Check the 12 h hold after the last XCR coat has passed (CS-004 §7.4 step 1).
2. Wipe on a thin film of Formula Five wax.
3. Let it haze.
4. Buff with a clean microfiber.
5. Repeat for 4 coats minimum on a new mold. The 4 is the team's count; the Formula Five guide asks for at least three on a new mold (`03 Datasheets/FORMULA-FIVE-Mold-Release-Wax.pdf`).
6. Wait at least 1 h after the last wax coat, so its solvents out-gas before the PVA (CS-004 §7.4 step 2).
7. Apply Partall #10 PVA in thin, even coats (CS-004 §7.4 step 3).
8. Let each coat dry fully to a glossy film. The Partall guide gives about 15–45 minutes a coat, depending on temperature, humidity and thickness (CS-004 §7.4 step 3).
9. Two coats. The 2 is the team's count; the Partall guide gives a mist coat then flow coats on a new mold.

Re-used sealed mold: clean with Clean 'N Glaze, then 1–2 wax coats, the 1 h wait and PVA as above (CS-004 §7.4 step 4).

Check: continuous PVA gloss with no fisheyes or dry patches (CS-004 §8). A dull, hazy or grainy PVA film can mean the coat is too thin (Partall guide).

8. When is the mold Ready for layup? Recording and labeling it

A mold is ready when it is sealed, released, recorded and labeled, and its Stage is set to Ready for layup. Then it is cleared for infusion (CS-006) or wet layup (CS-007) (CS-004 §7.5 step 5).

Also called: ready for layup, mold label, mould label, sticker, Sealed on, Stage, mold record, identify.

1. On the Molds tab, open the mold (CS-004 §7.4 step 5).
2. Set Sealing system to XCR, Sealed on to the date of the last XCR coat, Sealed by to yourself, and Stage to Sealed.
3. In the "Seal and release mold" note, write the XCR batch masses, coat count and coat times, and the wax and PVA coat counts.

4. Pick a label patch on the back face, inboard of the 40 mm keep-out at every edge, so never within 40 mm of the flange perimeter (CS-004 §7.5 step 1).
5. Scuff the patch to 220 grit and wipe with IPA (CS-004 §7.5 step 2).
6. Print the mold's label from the app and apply it. Check its parts-pulled count, seal date and sealed-by name (CS-004 §7.5 step 3).
7. Check the engraved mold ID is still legible; re-engrave or stamp it if not (CS-004 §7.5 step 4).
8. Set Stage to Ready for layup (CS-004 §7.5 step 5).

Labeling is last on purpose: a label put on before the XCR gets brushed over or peeled off, and a label edge under the seal is a leak channel you will not find at 27 inHg (CS-004 §7.5). On a re-seal, print a new label with the new seal date and the incremented parts-pulled count.